Question #1 of 43

Question ID: 1674742

Which of the following is *least accurate* of the use of mean-variance optimization as an asset allocation tool for institutional investors that incorporate illiquid assets?

A) Mean-variance optimization falls short as an adequate asset allocation tool for many illiquid asset classes.

B) Single-period mean-variance optimization fails when illiquid asset classes are introduced.

C) Mean-variance optimization works effectively if hedge fund return series data is smoothed before determining a covariance matrix in the MVO process.

Question #2 of 43

Question ID: 1674722

In a private equity fund, capital calls:

A) occur according to a predetermined schedule.

B) are required to occur within three years of the commitment date by mutual agreement.

C) may limit the rate of return an investor earns on the private equity fund.

Question #3 of 43

Question ID: 1674724

A group of analysts working in the hedge fund industry are having a discussion on liquidity risks associated with investing in hedge funds. They make the following comments:

Analyst 1:	Long-only, equity-oriented hedge funds are mostly more liquid compared to funds that make greater use of short selling.
Analyst 2:	A levered fund may be subject to higher liquidity risk when markets are stressed.
Analyst 3:	Event-driven strategies, by their nature, tend to have shorter investment horizons than other hedge fund strategies.

Which of the comments are correct?

A) Analyst 1 and Analyst 2 are correct.

B) Only Analyst 1 is correct.

C) Analyst 1 and Analyst 3 are correct.

Question #4 of 43

Question ID: 1674726

In the context of a private equity fund structured as a limited partnership, which of the following items increases available liquidity from the perspective of the limited partners?

A) Rebalancing.

B) A distribution.

C) A capital call.

An endowment fund has a current allocation of 70% equity and 30% fixed income. The fund is considering reallocating some of the equity and fixed income to alternative investments to better position the fund, given the current economic environment and to meet long-term return goals. Among the investments under consideration are private equity, commodities (through futures contracts), private real estate, distressed debt, and REITs, as well as adjustments to the 70/30 allocation without adding alternative investments.

Question #5 - 8 of 43

Question ID: 1674716

Assuming the primary focus of the fund is to reduce the risk of not achieving long-term return goals, which of the following investments is generally *most appropriate*?

A) A new allocation to commodities.

B) An increased allocation to fixed income.

C) A new allocation to private equity.

Question #6 - 8 of 43

Question ID: 1674717

A director of the fund has expressed a concern that the risk of one of the alternative investments may be understated. To which of the following investments is that concern *most likely* related?

A) Commodities.

B) Private real estate.

C) REITs.

Question #7 - 8 of 43

Question ID: 1674718

Assume the outlook for the economic cycle includes a forecast of high nominal economic growth, but only moderate real growth. An increased allocation to which of the following would be *best*?

A) Private equity.

B) Nonindexed bonds.

C) Commodities.

Question #8 - 8 of 43

Question ID: 1674719

Assume the fund begins to use a risk factor approach to selecting investments where the fund desires protection against inflation, but also desires liquidity. Given that approach, an addition of which of the following is *best*?

A) REITs.

B) Private real estate.

C) Distressed debt.

Question #9 of 43

Question ID: 1674728

Paul Wood is a high-net-worth individual with a long investment horizon. He believes that markets are fundamentally efficient. His advisor recommends an investment in hedge funds to meet Wood's requirement of high returns. This would be the first alternative investment made by the investor. In recommending hedge funds to Wood, the advisor has *most likely* breached suitability considerations with respect to:

A) governance.

B) investment horizon.

C) expertise.

Question #10 of 43

Question ID: 1674725

Which of the following *most likely* represents the compensation to a sponsor of a private equity fund?

A) A management fee of 2% and an incentive fee of 20%.

B) A management fee of 2% and an incentive fee of 2%.

C) A management fee of 10% and an incentive fee of 10%.

A pension fund is considering alternative investments as part of its asset allocation. In doing so, three analysts have commented on the process. Manuel states that distributions with positive excess kurtosis can be properly accounted for by using two market environments and weighting them appropriately with a probability for each environment. Nelson disagrees and expresses concern that combining two distributions is not appropriate, but he suggests a Monte Carlo simulation because fat tails can be accounted for in that type of analysis. Oliver is insistent that mean-variance optimization without constraints should not be used

because it could result in an overallocation to alternative investments that are not traded frequently.

Question #11 - 13 of 43

Question ID: 1674737

Regarding the statement by Manuel, which of the following is *most accurate*?

A) Manuel is proposing a correct technique for incorporating fat-tail distributions.

B) Manuel is proposing an incorrect technique, as two distributions combined will be normal.

C) Manuel is proposing an incorrect technique because it is unlikely for alternative investments to have positive excess kurtosis due to diversification.

Question #12 - 13 of 43

Question ID: 1674738

Nelson's comments are:

A) incorrect about the Monte Carlo simulation, but correct about combining distributions.

B) incorrect about both the Monte Carlo simulation and about combining distributions.

C) correct about the Monte Carlo simulation, but incorrect about combining distributions.

Question #13 - 13 of 43

Question ID: 1674739

Oliver's statement about mean-variance optimization is:

A) correct that mean-variance optimization would result in an overallocation for nonfrequently traded assets.

B) incorrect because nonfrequently traded assets would likely have a higher variance and a lower allocation.

C) incorrect because mean-variance optimization cannot be used with alternative investments due to transactions costs.

Question #14 of 43

Question ID: 1674729

Alba Ameen, CFA, manages an alternative asset portfolio at Jarus Investments. The lack of transparency in alternative assets causes challenges in reporting portfolio performance. To address this, Ameen has established the following control mechanisms:

I. Appointing independent administrators
II. Appointing an independent and respected accounting firm to perform the audit function
III. Allowing wide discretion in the valuation of fund assets

Which of the mechanisms is *least appropriate*?

A) Mechanism II.

B) Mechanism I.

C) Mechanism III.

Question #15 of 43

Question ID: 1674740

Risk factor–based asset allocation provides robust asset allocations, but care must be taken because of the following:

I. Investors may hold different definitions for a given risk factor.
II. Correlations among risk factors may shift under changing market conditions.
III. Factor sensitives are generally very stable.

Which of the following is *most accurate*?

A) Statements I and III are correct.

B) Statements I and II are correct.

C) Statements II and III are correct.

Question #16 of 43

Question ID: 1674712

An endowment wishing to diversify equity market risk is considering adding either bonds or alternative investments to its portfolio. A disadvantage of using bonds instead of an alternative asset is that:

A) changes in inflation levels could adversely impact the correlation of stocks and bonds.

B) bonds tend to have a negative correlation with equities.

C) bonds have higher expected returns than alternative assets over the long term.

Question #17 of 43

Question ID: 1674746

Which of the following concerns about monitoring alternative investments is *most* related to the wide range of discretion that managers of alternative investment funds have in selecting investments?

A) Custodian services.

B) Style drift.

C) Creation of a new fund by the current manager.

Question #18 of 43

Question ID: 1674709

Mark Jansen is a private credit analyst at Rockfort Asset Management. He has been asked by his supervisor to make a presentation on the role of direct lending and distressed investment in a portfolio. In his presentation to his supervisors, he makes the following statements:

- Both strategies involve the ownership of fixed-income assets.
- Distressed debt investments act more like fixed income, while direct lending investments act more like equity.
- Illiquidity risks are high with both strategies.

Which statement made by Jansen is *least accurate*?

A) Distressed debt investments act more like fixed income, while direct lending investments act more like equity.

B) Both strategies involve the ownership of fixed-income assets.

C) Illiquidity risks are high with both strategies.

Question #19 of 43

Question ID: 1674747

A limited partner in a private alternative investment fund observes that a general partner has withdrawn a significant portion of their own capital from the fund. Such an observation would raise a red flag with respect to:

A) style drift.

B) key person risk.

C) alignment of interests.

Question #20 of 43

Question ID: 1674708

The Xanuz Foundation (Xanuz) has historically invested solely in traditional asset classes. A recent review by the investment committee has recommended including alternative investments in the portfolio for the first time. The research committee of Xanuz is forecasting a high probability of a major market shock occurring over the next few quarters. Given this forecast, the *most appropriate* alternative asset class to add to the portfolio is:

A) gold.

B) private equity.

C) bonds.

Question #21 of 43

Question ID: 1674720

A hedge fund employs a side pocket as part of its structure. This feature will *most likely* result in which of the following?

A) Liquidity that is overestimated due to restrictions on redemptions.

B) Risk that is overestimated due to lower standard deviations from illiquid investments.

C) Returns that are overestimated due to capital calls being less than committed capital.

Question #22 of 43

Question ID: 1674707

Which of the following *best* represents an income-producing investment with limited liquidity?

A) Distressed debt.

B) Private equity.

C) Direct lending.

Question #23 of 43

Question ID: 1674743

An investor in a private equity fund who wishes to maintain their target asset allocation should *most likely* allocate undrawn capital to:

A) REITs.

B) public equity.

C) cash.

Question #24 of 43

Question ID: 1674748

Private equity, credit, and real estate returns are typically reported using internal rates of return (IRRs) rather than time-weighted returns (TWR). However, IRRs are sensitive to the timing of cash flows into and out of the fund. What other measures can be used in the performance evaluation of alternative investments?

A) Multiple on invested capital only.

B) Both multiple on invested capital and peer group analysis.

C) Peer group analysis only.

Question #25 of 43

Question ID: 1674744

An endowment fund has a target asset allocation of 10% in private equity. To achieve that allocation, the fund sets aside 10% of assets to meet capital calls as they occur. The 10% should ideally be invested in:

A) the same allocation the fund currently has in the remaining 90% not invested in private equity.

B) liquid investments similar to those expected when the private equity capital calls occur.

C) cash to meet the capital calls.

Question #26 of 43

Question ID: 1674727

The TJ Fund is an endowment fund guided by directors who believe that markets are not price efficient. They also believe in the principles of taking moderate risk and in complete transparency. Which of the following *best* represents why the fund would consider alternative investments?

A) The need for complete transparency.

B) The accepting of moderate risk.

C) The belief in a market that is not price efficient.

Question #27 of 43

Question ID: 1674710

Which of the following investments is *least likely* to be considered a form of indirect investment in commodities?

A) Long position in coffee futures expiring three months from now.

B) Investment in a publicly traded mutual fund that invests in agriculture.

C) Long futures position on the shares of a commodity trading company.

Candace Johnson, CFA, is not happy. Stock and bond markets have both decreased in value significantly over the last year, and her employer, Perpetual Investments, Inc. (Perpetual), has seen a lot of clients close their accounts. Johnson has been urging the investment committee to consider investing in alternative assets. She feels that alternative assets will

improve Perpetual's investment performance and possibly reduce the number of redemptions.

Johnson prepares a report for the investment committee. She makes three statements regarding various classes of alternative investments:

Statement 1:	Some hedge funds reduce overall portfolio beta.
Statement 2:	Private equity provides excellent diversification from public equity and fixed-income investments.
Statement 3:	Direct lending investments are good sources of income.

Johnson believes that certain alternative assets can serve as a portfolio risk mitigator. The committee is unsure about that and asks Johnson to prepare a brief presentation supporting her belief.

In light of the recent redemptions, the committee is concerned that alternative assets may not be feasible investments because of their illiquid nature. Johnson assures the committee that initially, she will invest only in hedge funds that have large *side pockets*. She explains that those side pockets allow investors to withdraw funds in periods of market turmoil when liquidity is compromised.

Finally, the committee is uncertain as to how suitable alternative assets are for its current investor base. Johnson assures them that alternative assets are suitable for Perpetual's entire investment base. Perpetual's investor base can be categorized as follows:

Category 1:	Young—average age of 30
Category 2:	Entrepreneurial—high risk-takers
Category 3:	Analytical—very detail oriented

Question #28 - 31 of 43

Question ID: 1674731

Which statement in Johnson's report is *least accurate*?

A) Statement 3.

B) Statement 2.

C) Statement 1.

Question #29 - 31 of 43

Question ID: 1674732

Johnson's presentation to the committee regarding risk mitigation should recommend the inclusion of which alternative asset class?

A) Private credit.

B) Real assets.

C) Private equity.

Question #30 - 31 of 43

Question ID: 1674733

Johnson's response to the committee regarding the side pocket is:

A) incorrect—large side pockets increase liquidity risk.

B) correct.

C) incorrect—side pockets are not a liquidity consideration.

Question #31 - 31 of 43

Question ID: 1674734

Which category within Perpetual's investor base would be *least* suitable to invest in alternative assets?

A) Category 2.

B) Category 1.

C) Category 3.

Question #32 of 43

Question ID: 1674735

In analyzing alternative investments, a test of serial correlation may be performed. The purpose of this test is to:

A) assess whether certain macroeconomic factors affect the correlated returns.

B) understand if the risk of the alternative investment is understated.

C) determine a profit strategy.

Question #33 of 43

Question ID: 1674741

Jonathan Green is using Monte Carlo simulation to identify the appropriate asset allocation for his client. What is the first step in the process?

A) Generating scenarios using the different means and covariances estimated under the different regimes.

B) Identifying variables to be randomly generated as part of the simulation process.

C) Simulating skewed and fat-tailed financial variables by estimating the behavior of factors and/or assets in low-volatility regimes and high-volatility regimes.

Question #34 of 43

Question ID: 1674723

An investor is looking to invest in an alternative asset class through funds-of-funds (FoFs). Which of the following features of a FoF is *least likely* to support their use by the investor versus a direct investment in individual funds?

A) The level of diversification across the asset class.

B) The level of outsourced expertise in fund manager due diligence.

C) The degree of flexibility to tailor exposure.

Adrianna Payne is the chief investment officer (CIO) of the Higher Education University endowment fund (HEU). Payne is concerned that recent geopolitical events and supply-side shocks—resulting in rising inflation, rising bond yields, and high equity market volatility—are having a negative impact on HEU's performance. It particular, Payne is focused on two risk objectives:

1. Diversify short-term equity market volatility in the portfolio.
2. Prevent a reduction in the probability of meeting the long-term mission of the endowment.

Payne meets with investment consultant Garth Bridge regarding the asset allocation of the endowment. Bridge recommends that HEU increases its allocation to government bonds.

Payne disagrees with Bridge that an increased allocation to government bonds will help meet either of the two risk objectives in the current macroeconomic environment.

Payne instead resolves to increase the allocation of the portfolio to alternative investments. She asks Bridge which alternative assets should be considered given her concerns about interest rate risk and inflation risk. Bridge provides Payne with a risk-factor sensitivity report for both traditional and alternative asset classes based on data from the past two decades. An extract from this report is displayed in **Exhibit 1: Risk Factor Sensitivity Report Extract** (note the report only shows risk factor sensitivities where they are statistically significant).

Exhibit 1: Risk Factor Sensitivity Report Extract

	Risk Factor	
Asset Class	**Nominal Duration**	**Inflation**
U.S. equities		
Government bonds	5.2	
Inflation-linked bonds	6.5	8.2
Commodities		12.5
Public real estate	4.5	1.3
Hedge funds		

Payne is concerned about the impact of an increased allocation to alternative investments on portfolio liquidity, particularly from exposure to private limited partnership funds. Bridge suggests Payne engages in liquidity planning using the well-known model of Takahashi and Alexander (2001). He demonstrates this model using the following theoretical example for a private fund allocation in its fourth year:

- Total committed capital = EUR 1,000
- Rate of contribution in Year 4 = 35%
- Capital already paid in before Year 4 = EUR 600
- Rate of distribution in Year 4 = 25%

- NAV in Year 3 = EUR 662
- Annual assumed growth rate in NAV = 10%

Payne is also concerned about the increase in negative skewness and long tails introduced into the return distribution of the portfolio through increasing the allocation to alternative investments. She asks Bridge which portfolio optimization techniques best accommodate such concerns.

Question #35 - 38 of 43

Question ID: 1674750

Payne's response to Bridge's asset allocation recommendation is:

A) correct regarding Risk Objective 2, but incorrect regarding Risk Objective 1.

B) correct regarding Risk Objective 1, but incorrect regarding Risk Objective 2.

C) correct.

Question #36 - 38 of 43

Question ID: 1674751

Based only on the data in Exhibit 1, the *most appropriate* asset class in addressing Payne's concerns of interest rate risk and inflation risk is:

A) inflation-linked bonds.

B) hedge funds.

C) commodities.

Question #37 - 38 of 43

Question ID: 1674752

Based on the theoretical liquidity planning data provided by Bridge and using the Takahashi and Alexander model, the net liquidity position (distributions – contributions) for Year 4 is *closest* to:

A) –EUR 42.

B) + EUR 42.

C) + EUR 100.

Question #38 - 38 of 43

Question ID: 1674753

In response to Payne's concerns regarding negative skewness and long tails, Bridge should recommend using which of the following asset allocation methods?

A) Mean-variance optimization without constraints.

B) Mean-variance optimization with constraints.

C) Mean-CVaR optimization.

Question #39 of 43

Question ID: 1674745

Which of the following is *most correct* regarding returns and reporting for alternative investments?

A) Stale pricing is not generally a concern because investments are selected opportunistically, and such pricing is accepted as part of the fund's strategy.

B) Significant delays in reporting can present a challenge for monitoring alternative investment funds.

C) Alternative investment firms use time-weighted rates of return in reporting.

Question #40 of 43

Question ID: 1674721

A disadvantage of undertakings for collective investments in transferable securities (UCITS) compared to traditional limited partnerships is:

A) limited investment choice due to regulation.

B) low liquidity.

C) returns that are fat tailed.

Question #41 of 43

Question ID: 1674714

A disadvantage of using a risk factor–based asset allocation approach relative to a traditional approach is:

A) practical hurdles in implementing portfolio mandates.

B) overestimation of diversification.

C) ambiguous primary drivers of risk.

Question #42 of 43

Question ID: 1674711

When considering the risk-reduction properties of alternative assets, it is sometimes noted that the reported returns of alternative assets exhibit lower volatility than the returns of public equities. The observation would be correct if:

A) there is strong positive correlation between the alternative asset returns and traditional asset returns.

B) the alternative asset returns exhibit serial correlation.

C) the alternative asset returns come from an index that has an absence of survivorship bias.

Question #43 of 43

Question ID: 1674713

Emanuel Seig, portfolio manager for HGI Funds, has recently attended a seminar about the impact of inflation on asset class performance. In his summary notes, he makes the following points:

1. With moderate inflation and high growth in an economy, high-yield bonds are expected to underperform.
2. With high inflation and low growth in the economy, inflation-linked bonds are expected to outperform.
3. Commodities are expected to outperform in a recessionary and deflationary environment.

With respect to the points written in his notes, it is *most likely* that:

A) none of the points are correct.

B) only two points are correct.

C) only one point is correct.